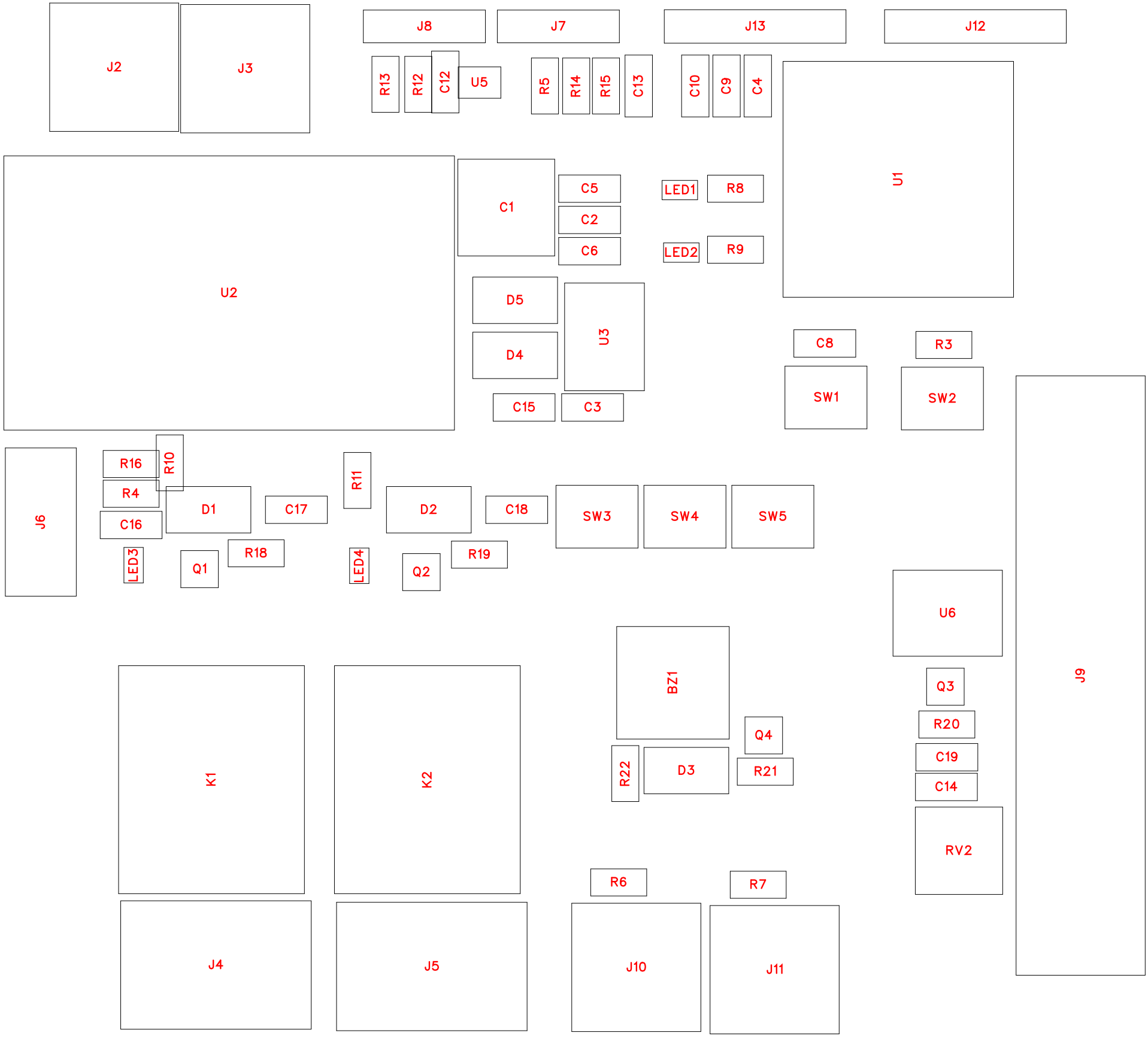
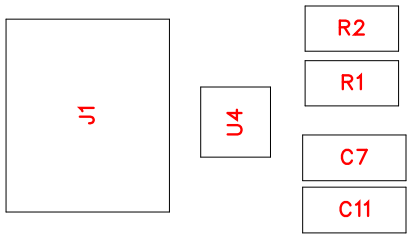
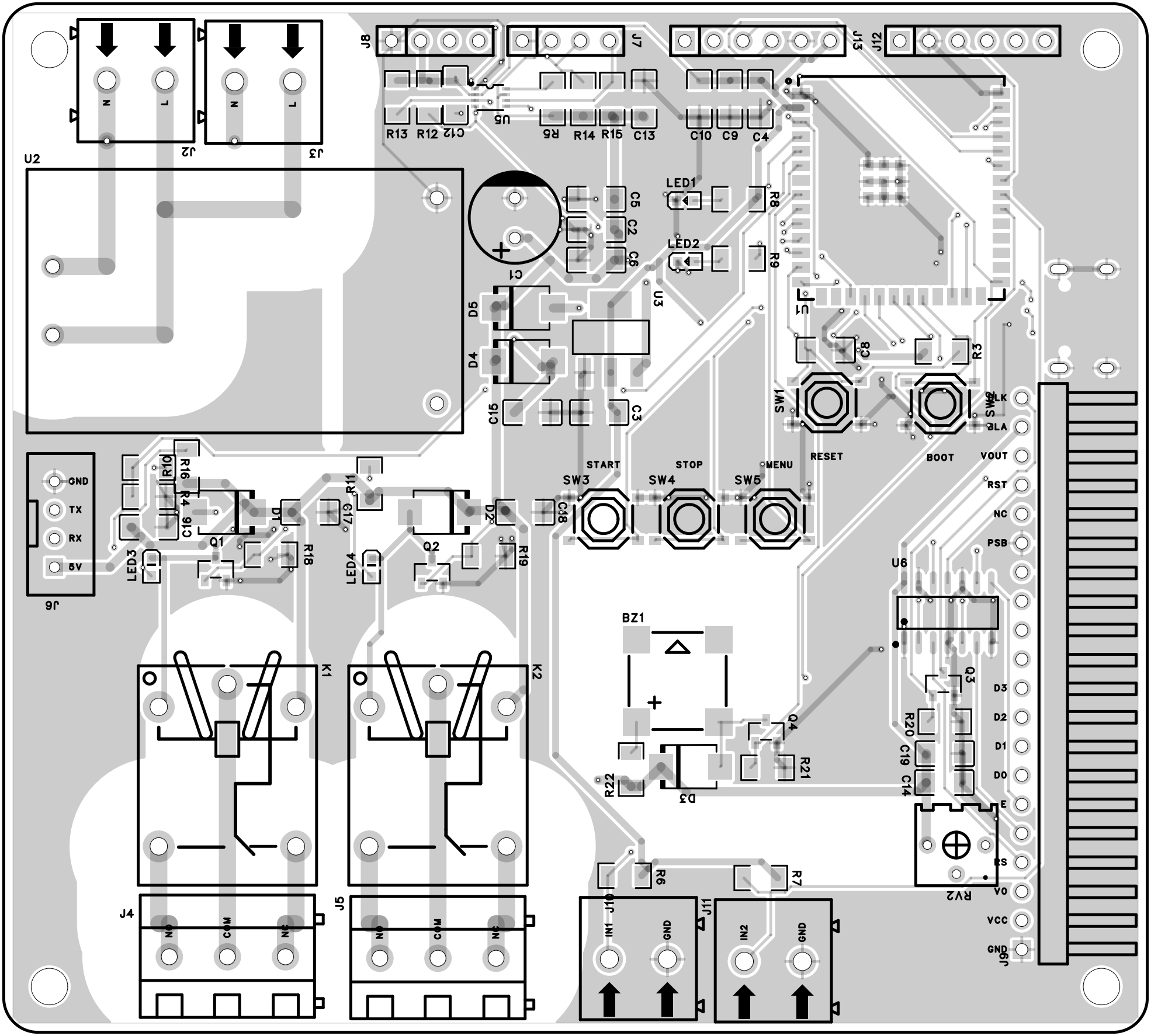
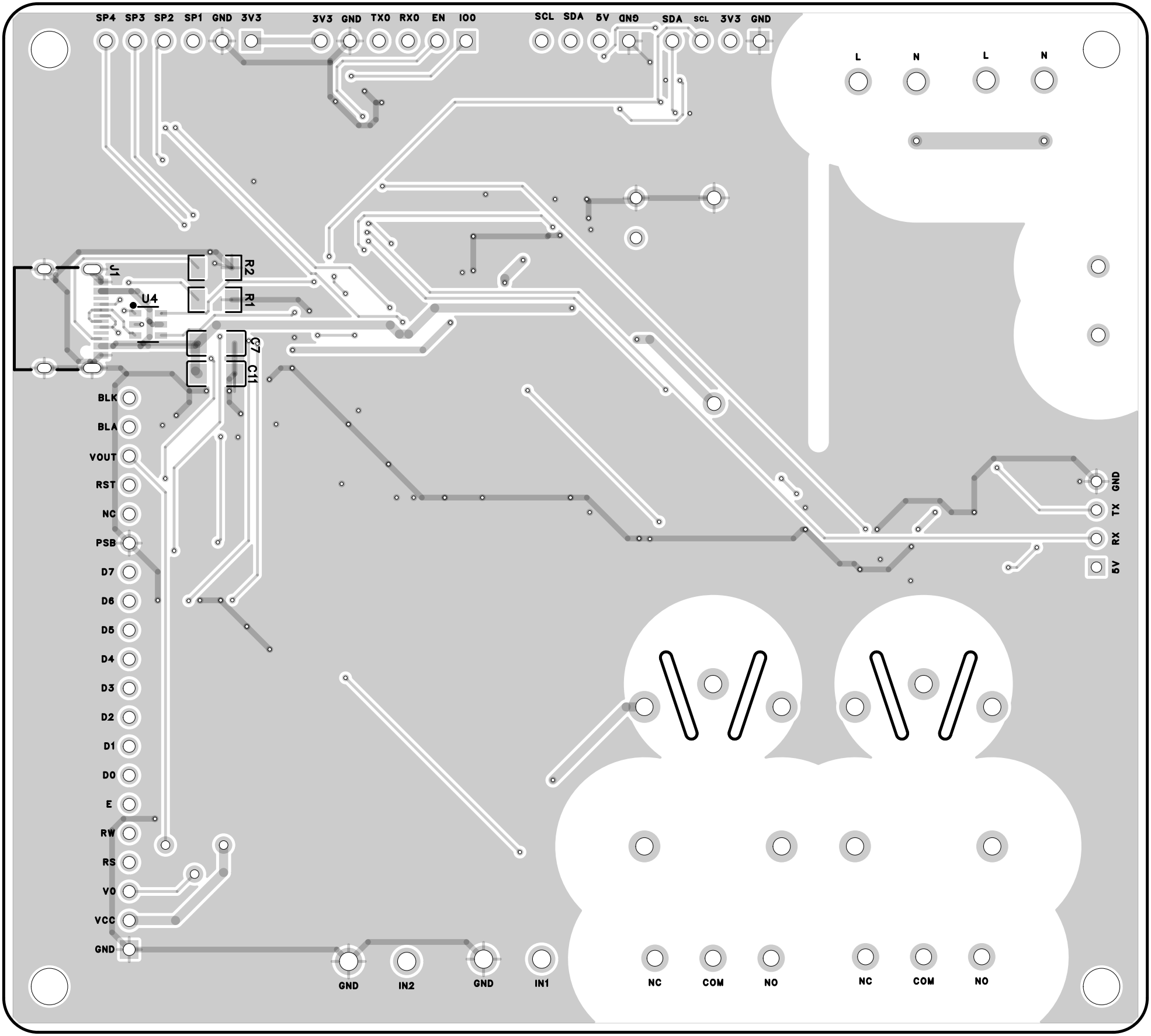

J2
J3
J8
J7
J13
J12
R13
R12
C12
U5
R5
R14
R15
C13
C10
C9
C4
U1
U2
C1
C5
C2
C6
LED1
R8
LED2
R9
D5
D4
U3
C15
C3
C8
R3
SW1
SW2
J6
R16
R10
R4
C16
D1
C17
R11
D2
C18
SW3
SW4
SW5
LED3
Q1
R18
LED4
Q2
R19
U6
J9
BZ1
Q3
R20
C19
C14
Q4
R22
D3
R21
RV2
K1
K2
R6
R7
J4
J5
J10
J11


J1
U4
R2
R1
C7
C11

No.	Quantity	Comment	Designator	Footprint	Value	Manufacturer Part	Manufacturer	Supplier Part	Supplier
1	1	2.7kHz	BZ1	BUZ-SMD_4P-L8.5-W8.5-P8.50-BR	2.7kHz	MLT-8530	华能	C94599	LCSC
2	1	470uF	C1	CAP-TH_BD8.0-P3.50-D0.6-FD-2	470uF	ERG16V470M8X12	JIERR(捷而瑞)	C47344174	LCSC
3	6	10uF	C2,C3,C4,C5,C6,C7	C1206	10uF	CL31A106KAHNNNE	SAMSUNG(三星)	C9807	LCSC
4	12	100nF	C8,C9,C10,C11,C12,C13,C14,C15,C16,C17,C18,C19	C1206	100nF	CL31B104KBCNNNC	SAMSUNG(三星)	C24497	LCSC
5	5	UMW SS34B	D1,D2,D3,D4,D5	SMB_L4.3-W3.6-LS5.4-RD		UMW SS34B	UMW(友台半导体)	C2758511	LCSC
6	1	TYPE-C-31-M-12	J1	USB-C_SMD-TYPE-C-31-M-12_1		TYPE-C-31-M-12	韩国韩荣	C165948	LCSC
7	4	KF128-5.08-2P-AA	J2,J3,J10,J11	CONN-TH_P5.08_KF128-5.08-2P		KF128-5.08-2P-AA	KEFA(科发)	C474952	LCSC
8	2	KF128-5.08-3P-AA	J4,J5	CONN-TH_3P-P5.08_KF128-5.08-3P		KF128-5.08-3P-AA	KEFA(科发)	C474953	LCSC
9	1	B4B-XH-A-G	J6	CONN-TH_B4B-XH-A-G		B4B-XH-A-G	JST	C594232	LCSC
10	2	PZ254V-11-04P	J7,J8	HDR-TH_4P-P2.54-V-M		PZ254V-11-04P	XFCN(兴飞)	C2691448	LCSC
11	1	PZ254R-11-20P	J9	HDR-TH_20P-P2.54-H-M-W9.7		PZ254R-11-20P	XFCN(兴飞)	C41413628	LCSC
12	2	PZ254V-11-06P	J12,J13	HDR-TH_6P-P2.54-V-M		PZ254V-11-06P	XFCN(兴飞)	C492405	LCSC
13	2	SRD-05VDC-SL-C	K1,K2	RELAY-TH_SRD-XXVDC-XL-C		SRD-05VDC-SL-C	松乐	C35449	LCSC
14	2	XL-1608SURC-04	LED1,LED2	LED0603-RD		XL-1608SURC-04	XINGLIGHT(成兴光)	C965798	LCSC
15	2	19-217/GHC-YR1S2/6T	LED3,LED4	LED0603-RD_GREEN		19-217/GHC-YR1S2/6T	EVERLIGHT(亿光)	C2986059	LCSC
16	4	AO3400A	Q1,Q2,Q3,Q4	SOT-23-3_L2.9-W1.3-P1.90-LS2.4-BR		AO3400A	AOS	C20917	LCSC
17	2	5.1kΩ	R1,R2	R1206	5.1kΩ	1206W4F5101T5E	UNI-ROYAL(厚声)	C26033	LCSC
18	5	10kΩ	R3,R4,R5,R6,R7	R1206	10kΩ	1206W4F1002T5E	UNI-ROYAL(厚声)	C17902	LCSC
19	4	1kΩ	R8,R9,R10,R11	R1206	1kΩ	1206W4F1001T5E	UNI-ROYAL(厚声)	C4410	LCSC

20	5	4.7kΩ	R12,R13,R14,R15,R16	R1206	4.7kΩ	1206W4F4701T5E	UNI-ROYAL(厚声)	C17936	LCSC
21	4	100kΩ	R18,R19,R20,R21	R1206	100kΩ	1206W4F1003T5E	UNI-ROYAL(厚声)	C17900	LCSC
22	1	10Ω	R22	R1206	10Ω	1206W4F100JT5E	UNI-ROYAL(厚声)	C17903	LCSC
23	1	10kΩ	RV2	RES-ADJ-TH_3362P	10kΩ	3362P-1-103LF	BOURNS	C58159	LCSC
24	5	TS-1187A-B-A-B	SW1,SW2,SW3,SW4,SW5	SW-SMD_4P-L5.1-W5.1-P3.70-LS6.5-TL_H1.5		TS-1187A-B-A-B	XKB Connection(中国星坤)	C318884	LCSC
25	1	2.4GHz	U1	WIRELM-SMD_ESP32-S3-WROOM-1U	2.4GHz	ESP32-S3-WROOM-1U-N16R8	ESPRESSIF(乐鑫)	C3013946	LCSC
26	1	HLK-5M05	U2	PWRM-TH_HLK-5M05		HLK-5M05	HI-LINK(海凌科)	C209907	LCSC
27	1	SGM2212-3.3XKC3G/TR	U3	SOT-223-3_L6.5-W3.4-P2.30-LS7.0-BR		SGM2212-3.3XKC3G/TR	SGMICRO(圣邦微)	C3294699	LCSC
28	1	USBLC6-2SC6	U4	SOT-23-6_L2.9-W1.6-P0.95-LS2.8-BL		USBLC6-2SC6	ST(意法半导体)	C7519	LCSC
29	1	PCA9306DCUR	U5	SSOP-8_L2.0-W2.3-P0.50-LS3.1-BR		PCA9306DCUR	TI(德州仪器)	C33196	LCSC
30	1	SN74AHCT125DR	U6	SOIC-14_L8.7-W3.9-P1.27-LS6.0-BL		SN74AHCT125DR	TI(德州仪器)	C155176	LCSC


J2
N
L
J3
N
L
J8
J7
J13
J12
U5
R13
R12
C12
R5
R14
R15
C13
C10
C9
C4
U2
LED1
R8
LED2
R9
C5
C2
C6
C1
U3
J1
D5
D4
C8
R3
C15
C3
SW1
SW2
CLK
DLA
START
STOP
MENU
RESET
BOOT
SW3
SW4
SW5
VOUT
RST
NC
PSB
GND
TX
RX
5V
J6
R16
R10
R4
C13
Q1
D1
R11
C17
D2
C18
LED3
LED4
Q2
R18
R19
U6
BZ1
Q3
Q4
R20
C19
C14
R21
R22
D3
D3
D2
D1
D0
E
RS
VO
VCC
GND
J9
RV2
K1
K2
J4
J5
NO
COM
NC
R6
R7
J10
J11
IN1
GND
IN2
GND


SP4
SP3
SP2
SP1
GND
3V3
3V3
GND
TX0
RX0
EN
IO0
SCL
SDA
5V
GND
SDA
SCL
3V3
GND
L
N
L
N
J1
U4
R2
R1
C7
C11
BLK
BLA
VOUT
RST
NC
PSB
D7
D6
D5
D4
D3
D2
D1
D0
E
RW
RS
VO
VCC
GND
GND
TX
RX
5V
GND
IN2
GND
IN1
NC
COM
NO
NC
COM
NO

